

Everyday AI sensemaking of freelancing knowledge workers

Lan Li

University of North Carolina at Chapel Hill
Chapel Hill, North Carolina, USA
ll12@unc.edu

ABSTRACT

In recent years, fast moving AI-enabled tools such as ChatGPT have tightly gripped society's imagination [7]. With them follow questions about AI's potential impact on the world of work. While debates and predictions have been brewing for some time, the discussion often pulsates between concerns of widespread worker displacement or the realization of a new workers' paradise. This study seeks to contextualize what working in the age of AI is like by examining the on the ground experience of knowledge workers, particularly those whose jobs are projected to hold high exposure to AI. It explores how they are making sense of, coping with, and adapting to this technological moment through their own vantage point. This paper outlines the preliminary findings from a qualitative pilot study of these workers' AI sensemaking process.

CCS CONCEPTS

• **Human-centered computing** → **Collaborative interaction**;
• **Empirical studies in HCI**; **HCI theory, concepts and models**;
• **Social and professional topics**;

KEYWORDS

knowledge worker, human-AI interaction, human-AI relational paradigm, sensemaking, skill set

ACM Reference Format:

Lan Li. 2023. Everyday AI sensemaking of freelancing knowledge workers. In *Computer Supported Cooperative Work and Social Computing (CSCW '23 Companion)*, October 14–18, 2023, Minneapolis, MN, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3584931.3608924>

1 INTRODUCTION

AI-enabled tools are increasingly being integrated into the context of work, particularly for knowledge workers [20]. Buoyed by the hype surrounding AI, and further fueled by the release of tools such as ChatGPT [17], within and outside of the work setting, a growing number of workers are themselves, in some cases with relative ease, coming into contact with and actively exploring the capabilities of AI-enabled tools; and in the process, potentially gaining greater awareness of and recognition towards their own interactions with these systems that might have been less visible before.

As AI-enabled tools continue to advance, at a seemingly astounding pace, in terms of its repertoire of abstract skills and capabilities,

in some cases purportedly exceeding that of human workers in certain limited task domains [12], from computer programmers [6] to translators [16], virtual assistants [8] to accountants [2], writers [4], lawyers [19], artists [18] and more, this rising level of awareness about AI also seems to usher in increasing concern, enthusiasm, as well as deep ambiguity and uncertainty regarding the potential of such technology to disrupt and to transform the nature of the work that knowledge workers engage in [9].

This state of uncertainty however does not seem to be alleviated by turning towards expert opinions as there appears to be little consensus amongst researchers regarding what AI's impact will be across occupations, industries, and the workers in those fields; and how it might influence long standing questions of job polarization and inequality [5].

Given the high degree of uncertainty about the potential impact of AI, coupled with growing ubiquity of direct interactions with these systems, the overarching questions driving this project are these: first, how do knowledge workers themselves make sense of AI and draw implications regarding AI's potential impact on their work? Second, what kind of relational paradigm do they develop towards AI through sensemaking and their direct engagement with these tools? In other words, how do they view 'AI-enabled tools' as? Is the relationship with AI, for example, competitive, augmentative, collaborative, a mix of these, or does the paradigm shift depending on the work context, and if so, how? Third, how does sensemaking, and more specifically, the relational paradigms they develop, shape future actions they take with respect to the type of work they engage in, the set of skills they choose to develop, and the career trajectories they hope to take? And lastly, what role does technology play in enabling and supporting knowledge workers' AI sensemaking process?

While occupationally-focused surveys have highlighted a variety of perceptions, predictions, and speculations knowledge workers hold toward AI, fewer empirical studies seem to explore the sensemaking processes knowledge workers undertake to arrive at those viewpoints [11]. This project seeks to contribute to this emerging understanding of knowledge workers' AI sensemaking process by capturing workers' own articulation of AI's impact on their livelihood and the everyday sociotechnical practices they engage in to cope with and adapt to the circumstances of this highly unpredictable moment.

2 COMPLETED WORK

So far, I have conducted an initial pilot study as a first step towards answering the research questions posed. In particular, this qualitative pilot study examines the AI sensemaking of 20 freelance knowledge workers on Upwork [15]. What makes these workers uniquely situated for understanding the sensemaking process they develop towards AI-enabled tools? There are two key factors.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CSCW '23 Companion, October 14–18, 2023, Minneapolis, MN, USA

© 2023 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0129-0/23/10.

<https://doi.org/10.1145/3584931.3608924>

First, as online freelancers, skill set plays a critical role in workers' continual acquisition and maintenance of work via the digital labor platform where they work [1]. This makes these workers ideal subjects to understand how AI is influencing the online labor market, skill demands, and how it is, seemingly in real time, shaping the nature of work for workers across a wide spectrum of skills and occupations while maintaining constant the platform context within which these diverse set of skills are being exercised [3].

Second, due to the wide spectrum of skill categories freelancing platforms support, the workers interviewed were further limited to those whose jobs may hold high exposure to AI [10]. The initial hypothesis was that, perhaps more salient than most, these workers may be faced with greater pressure to make sense of and adapt to their current situation. As such, the freelancers interviewed held jobs across eight skill categories. They are accountants and bookkeepers, data entry workers, editors and proofreaders, HR administrators, marketing researchers, translators, technical writers, and virtual assistants. During the interviews, these workers discussed the role AI-enabled tools play in their work practices, the relational paradigms they have developed towards AI, and projected future actions they might take given AI-enabled tools's continual development in their skill domain.

The data collection phase of this initial pilot is now complete and I am currently engaging in the data analysis and the development of a conceptual model to understand these workers' AI sensemaking process. Preliminary findings from the pilot study showed that while all of the participants were aware of advancements in AI, participants enacted varying degrees of sensemaking activities to understand AI and draw conclusions about its implications for their jobs.

But, a set of common sensemaking activities did emerge. These include, as of now, information discovery, verification, backcasting, and pattern recognition. In brief, information discovery speaks to both the active and passive means of learning about AI and its relationship to workers' jobs. Verification highlights both the reactive practices of fixing or repairing errors generated by AI-enabled tools as well as the proactive means of seeking out interactions with AI-enabled tools to better understand it and to improve their own workflow. Backcasting illustrates the synthesis or aggregation of workers' personal knowledge and experience about the nature of their work and how it might interface with AI. Finally, pattern recognition captures the ways in which workers attempt to track ongoing trends in their skill categories and connect the dots to anticipate how these trends might contribute to the shifting nature of their work.

Through engaging in these and other sensemaking activities, some workers have developed nuanced understanding regarding their relationship to AI-enabled tools. Rather than viewing AI in binary terms as purely competitive – echoing the sentiment of replacement by AI – or solely collaborative – echoing the sentiment of AI as a teammate, through engaging in these sensemaking activities, workers articulated aspects in which AI-enabled tools might be competitive, augmentative, as well as collaborative, in turn stitching together a far more nuanced understanding about AI-enabled tools and their relationship to them.

Regarding projected future actions, while few workers discussed undertaking major changes in their career due to AI, it is also not

the case that workers are not paying any attention whatsoever to new developments in AI. Most seemed to straddle between continuing their line of work as usual while also actively monitoring the ongoing progression of AI. This highlights the fact that while most do not feel as though AI will have much immediate impact on their work, many are nevertheless keeping an eye out towards its development through continual engagement in AI sensemaking.

Finally, the pilot also revealed that, as online freelancers, the design of the digital labor platform itself, despite its common characterization of precarity [13], did in various ways extend element of support towards workers' AI sensemaking process by enabling workers to acquire new skills through engaging with a diverse array of clients, assess which skills to develop by searching for jobs, and evolve their professional identity by meshing together multiple set of skills via their profile. Many of these platform affordances would have been more difficult to come by within the perhaps more well defined structure of an organization.

The pilot also brought into view that discussing the potential impact of AI on one's job can be a tricky and difficult conversation to hold, one that requires sensitivity and understanding. The pilot interview revealed the close tie between workers' skills and sense of identity [14]. While some workers seemed to be more hesitant about discussing this topic, many were open to discussing AI and implications for their work. Some have found the interview helpful in clarifying their own thinking around AI.

3 PROPOSED WORK

The pilot has been extremely helpful in shaping the sampling of workers and the continual refinement of the interview questions. For the dissertation project, as the current data collection process has not reached saturation point yet, I hope to extend the initial pilot to increase the number of workers interviewed.

The development of this AI sensemaking framework seeks to make contributions to the future design of tools and platforms workers use, relating to the sociotechnical core of the CSCW community. It may act as a roadmap towards better understanding how human-AI systems can be designed by enabling a complex set of human-AI relational paradigms that would ultimately empower workers. In addition, by uncovering agentic means through which knowledge workers are able to adapt to technological changes brought on by AI, this framework may offer potential training or support to enable workers to better cope with current and ongoing changes.

4 CONCLUSION

In summary, through the conceptual development of worker's AI sensemaking, this project seeks to clarify how knowledge workers themselves make sense of the current technological moment, what information ecology do they inhabit, how are they getting to know AI, what have they learned about these systems, and how does AI sensemaking and workers' sociotechnical practices shape the decisions they make regarding their work now and into the future. At this stage of my dissertation development process, participation in this doctoral consortium would enable me to present the working conceptual model of AI sensemaking I have developed so far through the pilot study and receive critical and timely feedback

regarding how the project can be further expanded for the next phase of data collection.

REFERENCES

- [1] Allie Blaising, Yasmine Kotturi, Chinmay Kulkarni, and Laura Dabbish. 2021. Making it Work, or Not: A Longitudinal Study of Career Trajectories Among Online Freelancers. *Proceedings of the ACM on Human-Computer Interaction* 4, CSCW3 (2021), 1–29.
- [2] Gary Bolinger. 2017. Artificial Intelligence and the Future of the Accounting Profession. <https://www.cpapracticeadvisor.com/2017/08/14/artificial-intelligence-and-the-future-of-the-accounting-profession/26543/>.
- [3] Fabian Braesemann, Fabian Stephany, Ole Teutloff, Otto Kässi, Mark Graham, and Vili Lehdonvirta. 2022. The global polarisation of remote work. *Plos one* 17, 10 (2022), e0274630.
- [4] Annie Brown. 2021. AI Is Not Going To Replace Writers Anytime Soon – But The Future Might Be Closer Than You Think. <https://www.forbes.com/sites/anniebrown/2021/07/20/ai-is-not-going-to-replace-writers-anytime-soon-but-the-future-might-be-closer-than-you-think/?sh=7dee35f56087>.
- [5] Erik Brynjolfsson, Tom Mitchell, and Daniel Rock. 2018. What can machines learn, and what does it mean for occupations and the economy?. In *AEA papers and proceedings*, Vol. 108. 43–47.
- [6] Davide Castelvetti. 2022. Are ChatGPT and AlphaCode going to replace programmers? *Nature* (2022).
- [7] Megan Cerullo. 2023. These 20 jobs are the most “exposed” to AI, ChatGPT, researchers say. <https://www.cbsnews.com/news/jobs-chatgpt-occupations-industries-exposed-teacher-telemarketer/>.
- [8] Brian X. Chen. 2023. How ChatGPT and Bard Performed as My Executive Assistants. <https://www.nytimes.com/2023/03/29/technology/personaltech/ai-chatgpt-google-bard-assistant.html>.
- [9] Lydia DePillis and Steve Lohr. 2023. Tinkering With ChatGPT, Workers Wonder: Will This Take My Job? <https://www.nytimes.com/2023/03/28/business/economy/jobs-ai-artificial-intelligence-chatgpt.html>.
- [10] Carl Benedikt Frey and Michael A Osborne. 2017. The future of employment: How susceptible are jobs to computerisation? *Technological forecasting and social change* 114 (2017), 254–280.
- [11] Masashi Goto. 2022. Accepting the future as ever-changing: professionals’ sense-making about artificial intelligence. *Journal of Professions and Organization* 9, 1 (2022), 77–99.
- [12] Ahmed Hosny, Chintan Parmar, John Quackenbush, Lawrence H Schwartz, and Hugo JWL Aerts. 2018. Artificial intelligence in radiology. *Nature Reviews Cancer* 18, 8 (2018), 500–510.
- [13] Mohammad Hossein Jarrahi, Will Sutherland, Sarah Beth Nelson, and Steve Sawyer. 2020. Platformic management, boundary resources for gig work, and worker autonomy. *Computer supported cooperative work (CSCW)* 29, 1 (2020), 153–189.
- [14] Arne L Kalleberg. 2009. Precarious work, insecure workers: Employment relations in transition. *American sociological review* 74, 1 (2009), 1–22.
- [15] Eliscia Kinder, Mohammad Hossein Jarrahi, and Will Sutherland. 2019. Gig Platforms, Tensions, Alliances and Ecosystems: An Actor-Network Perspective. *Proceedings of the ACM on Human-Computer Interaction* 3, CSCW (2019), 1–26.
- [16] Bernard Marr. 2018. Will Machine Learning AI Make Human Translators An Endangered Species? <https://www.forbes.com/sites/bernardmarr/2018/08/24/will-machine-learning-ai-make-human-translators-an-endangered-species/?sh=7fde723a3902>.
- [17] Cade Metz. 2020. Meet GPT-3. It has learned to code (and blog and argue). *The New York Times* 24 (2020).
- [18] Krzysztof Pelc. 2023. AI Will Make Human Art More Valuable. <https://www.wired.com/story/art-artificial-intelligence-history/>.
- [19] Chris Stokel-Walker. 2023. Generative AI Is Coming For the Lawyers. <https://www.wired.com/story/chatgpt-generative-ai-is-coming-for-the-lawyers/>.
- [20] Tom Warren. 2023. Microsoft to detail the future of work with AI during March 16th event. <https://www.theverge.com/2023/3/6/23627228/microsoft-ai-future-of-work-event-date>. (2023).